

6.1 (a) Show that $X(\Omega)$ is periodic with period 2π , that is, $X(\Omega + 2\pi) = X(\Omega)$.

(b) Specifically show that $X(\Omega) = \frac{1}{1 - ae^{-j\Omega}}$ is periodic.

F14122048 陳柏通
資訊116

(a)

$$X(\Omega + 2\pi) = \sum_{n=-\infty}^{\infty} x[n] e^{-j(\Omega + 2\pi)n}$$

$$= e^{-j2\pi n} \sum_{n=-\infty}^{\infty} x[n] e^{j\Omega n}$$

$$e^{-j2\pi n} = \sum_{n=-\infty}^{\infty} x[n] e^{-j2\pi n} = X(\Omega) \neq$$

$$= \cos(2\pi n) - j\sin(2\pi n)$$

$$= 1$$

$$(b) \quad X(\Omega + 2\pi) = \frac{1}{1 - ae^{j(\Omega + 2\pi)}}$$

$$= \frac{1}{1 - ae^{-j\Omega}}$$

$$= X(\Omega) \neq$$

6.9 A certain signal $x[n]$ has DTFT given by

$$X(\Omega) = \frac{e^{j\Omega}}{e^{j\Omega} + a}$$

where a is a constant. Find DTFT $Y(\Omega)$ of the following:

(a) $y[n] = x[n-2]$

(b) $y[n] = x[n]e^{j4n}$

(c) $y[n] = x[n]\sin 2n$

(d) $y[n] = x[n] * x[n]$

$$\begin{aligned} \text{(a)} \quad Y(\Omega) &= e^{-j2\Omega} X(\Omega) \\ &= e^{-j2\Omega} \cdot \frac{e^{j\Omega}}{e^{j\Omega} + a} \\ &= \frac{e^{-j\Omega}}{e^{j\Omega} + a} \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad Y(\Omega) &= X(\Omega - 4) \\ &= \frac{e^{j(\Omega - 4)}}{e^{j(\Omega - 4)} + a} \end{aligned}$$

$$\text{(c)} \quad y[n] = x[n] \frac{e^{j2n} - e^{-j2n}}{2j}$$

$$\begin{aligned} Y(\Omega) &= \frac{1}{2j} X(\Omega - 2) - \frac{1}{2j} X(\Omega + 2) \\ &= \frac{1}{2j} \frac{e^{j(\Omega - 2)}}{e^{j(\Omega - 2)} + a} - \frac{1}{2j} \frac{e^{j(\Omega + 2)}}{e^{j(\Omega + 2)} + a} \end{aligned}$$

$$\begin{aligned} \text{(d)} \quad Y(\Omega) &= X(\Omega) X(\Omega) \\ &= \left(\frac{e^{j\Omega}}{e^{j\Omega} + a} \right)^2 \\ &= \frac{e^{j2\Omega}}{(e^{j\Omega} + a)^2} \end{aligned}$$

6.13 Determine the signal $x[n]$ corresponding to each of the following Fourier transforms:

(a) $X(\Omega) = 1 + 3e^{-j2\Omega} - 2e^{j4\Omega} + e^{j5\Omega}$

(b) $X(\Omega) = \frac{e^{-j\Omega} - \frac{1}{2}}{1 - \frac{1}{2}e^{-j\Omega}}$

(c) $X(\Omega) = 3\pi[\delta(\Omega-2) + \delta(\Omega+2)]$

(a) $X(\Omega) = 1 + 3e^{-j2\Omega} - 2e^{j4\Omega} + e^{j5\Omega}$

$x[n] = \delta[n] + 3\delta[n-2] - 2\delta[n+4] + \delta[n+5]$ *

(b) $X(\Omega) = \frac{e^{-j\Omega}}{1 - \frac{1}{2}e^{-j\Omega}} - \frac{\frac{1}{2}}{1 - \frac{1}{2}e^{-j\Omega}}$

$$x[n] = \left(\frac{1}{2}\right)^{n-1} u[n-1] - \frac{1}{2} \left(\frac{1}{2}\right)^n u[n]$$

$$= \left(\frac{1}{2}\right)^{n-1} u[n-1] - \left(\frac{1}{2}\right)^{n+1} u[n] *$$

(c) $X(\Omega) = \frac{3}{2} [2\pi \delta(\Omega-2) + 2\pi \delta(\Omega+2)]$

$$x[n] = \frac{3}{2} (e^{j2n} + e^{-j2n})$$

$$= 3 \cos(2n) *$$

6.18 Find the convolution $y[n] = h[n] * x[n]$ of the following pairs of signals:

(a) $x[n] = \left(\frac{1}{4}\right)^n u[n]$, $h[n] = 1$

(b) $x[n] = \left(\frac{1}{3}\right)^n u[n]$, $h[n] = \delta[n] + \delta[n-1]$

(c) $x[n] = \left(\frac{1}{2}\right)^n u[n]$, $h[n] = \left(\frac{1}{3}\right)^n u[n]$

(a) $Y(\omega) = X(\omega) H(\omega)$

$$= \frac{2\pi \delta(\omega)}{1 - \frac{1}{4}e^{j\omega}}$$

$$= \frac{2\pi \delta(\omega)}{1 - \frac{1}{4}e^0}$$

$$= \frac{4}{3} \cdot 2\pi \delta(\omega)$$

$$y[n] = \frac{4}{3} \delta[n]$$

(b) $Y(\omega) = \frac{1}{1 - \frac{1}{3}e^{j\omega}} + \frac{e^{-j\omega}}{1 - \frac{1}{3}e^{-j\omega}}$

$$y[n] = \left(\frac{1}{3}\right)^n u[n] + \left(\frac{1}{3}\right)^{n-1} u[n-1] *$$

(c) $Y(\omega) = \frac{1}{(1 - \frac{1}{2}e^{j\omega})(1 - \frac{1}{3}e^{j\omega})}$

$$= \frac{3}{1 - \frac{1}{2}e^{j\omega}} - \frac{2}{1 - \frac{1}{3}e^{j\omega}}$$

$$y[n] = 3 \cdot \left(\frac{1}{2}\right)^n u[n] - 2 \cdot \left(\frac{1}{3}\right)^n u[n] *$$

6.19 Find the DFT of the sequence $x[n] = a^n$.

$$X[k] = \sum_{n=0}^{N-1} a^n (e^{j\frac{2\pi}{N}k})^n$$

$$= \sum_{n=0}^{N-1} (ae^{j\frac{2\pi}{N}k})^n$$

$$= \frac{1 - (ae^{j\frac{2\pi}{N}k})^N}{1 - ae^{j\frac{2\pi}{N}k}}$$

$$= \frac{1 - a^N}{1 - ae^{j\frac{2\pi}{N}k}} \neq$$

$$e^{j\pi k} = \cos(\pi k) - j\sin(\pi k), \quad k=0, 1, \dots, N-1$$

$$= 1$$

6.21 Find the DFT of the following sequences:

(a) $x[n] = \{0, 1, 2, 3\}$

(b) $y[n] = \{1, 1, -1, -1, 1, 1, -1, -1\}$

$$(a) X[k] = \sum_{n=0}^3 x[n] e^{-j\frac{\pi}{4}kn} \quad 0 \leq k \leq N-1$$

$$= e^{-j\frac{\pi}{2}k} + 2e^{-j\pi k} + 3e^{-j\frac{3\pi}{2}k}$$

$$X[0] = 1 + 2 + 3 = 6 \quad e^{j\frac{\pi}{2}} = j$$

$$X[1] = -j + 2(-1) + 3j \quad e^{-j\pi} = -1$$

$$= -2 + 2j \quad e^{-j\frac{3\pi}{2}} = j$$

$$X[2] = -1 + 2 - 3 = -2$$

$$X[3] = j - 2 - 3j = -2 - 2j$$

$$X[k] = \{6, -2 + 2j, -2, -2 - 2j\} \neq$$

(b) $x_0[n] = \{1, 1, -1, 1\}$

$$y[n] = x_0[n] + x_0[n-4]$$

$$Y[k] = X_0[k] + X_0[k] \cdot (-1)^k = X_0[k] (1 + (-1)^k)$$

$$X_0[k] = \sum_{n=0}^3 x_0[n] e^{-j\frac{2\pi}{8}kn} = 1 + e^{-j\frac{\pi}{4}k} - e^{-j\frac{\pi}{2}k} - e^{-j\frac{3\pi}{4}k} \quad 0 \leq k \leq N-1$$

$$Y[1] = Y[3] = Y[5] = Y[7] = 0$$

$$Y[0] = X_0[0] = 0$$

$$Y[2] = 2 \cdot X_0[2] = 2 \cdot (1 - j + 1 - j) = 4 - 4j$$

$$Y[4] = 2 \cdot X_0[4] = 2 \cdot (1 - 1 - 1 + 1) = 0$$

$$Y[6] = 2 \cdot X_0[6] = 4 + 4j$$

$$Y[k] = \{0, 0, 4 - 4j, 0, 0, 0, 4 + 4j, 0\} \neq$$

6.23 Show that

$$(a) \ x[n] \cos\left(\frac{2\pi mn}{N}\right) \longleftrightarrow \frac{1}{2} [X(k-m) + X(k+m)]$$

$$(b) \ x[n] \sin\left(\frac{2\pi mn}{N}\right) \longleftrightarrow \frac{1}{2j} [X(k-m) - X(k+m)]$$

$$\begin{aligned} (a) \ y[n] &= x[n] \cos\left(\frac{2\pi mn}{N}\right) \\ &= x[n] \left(\frac{e^{j\frac{2\pi}{N}mn} + e^{-j\frac{2\pi}{N}mn}}{2} \right) \\ &= \frac{1}{2} x[n] e^{j\frac{2\pi}{N}mn} + \frac{1}{2} x[n] e^{-j\frac{2\pi}{N}mn} \end{aligned}$$

$$Y[k] = \frac{1}{2} [X(k-m) + X(k+m)] \quad \#$$

$$\begin{aligned} (b) \ y[n] &= x[n] \sin\left(\frac{2\pi mn}{N}\right) \\ &= x[n] \left(\frac{e^{j\frac{2\pi}{N}mn} - e^{-j\frac{2\pi}{N}mn}}{2j} \right) \\ &= \frac{1}{2j} x[n] e^{j\frac{2\pi}{N}mn} - \frac{1}{2j} x[n] e^{-j\frac{2\pi}{N}mn} \end{aligned}$$

$$Y[k] = \frac{1}{2j} [X(k-m) - X(k+m)] \quad \#$$

$$\begin{aligned} & \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}kn} \cdot e^{j\frac{2\pi}{N}mn} \\ &= \sum_{n=0}^{N-1} x[n] e^{j\frac{2\pi}{N}n(k-m)} \\ &= X(k-m) \end{aligned}$$

6.25 The DFT of a signal $x[n]$ is

$$X(0)=1, \quad X(1)=1+j2, \quad X(2)=1-j, \quad X(3)=1+j, \quad X(4)=1-j2$$

Compute $x[n]$.

$$X[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j \frac{2\pi}{5} kn}, \quad 0 \leq n \leq N-1$$

$$= \frac{1}{5} \sum_{k=0}^4 X[k] e^{j \frac{2\pi}{5} kn}$$

$$x[n] = \frac{1}{5} \left(X[0] + X[1] e^{j \frac{2\pi}{5} n} + X[2] e^{j \frac{4\pi}{5} n} + X[3] e^{j \frac{6\pi}{5} n} + X[4] e^{j \frac{8\pi}{5} n} \right)$$

$$= \frac{1}{5} \left(1 + (1+j2) e^{j \frac{2\pi}{5} n} + (1-j) e^{j \frac{4\pi}{5} n} + (1+j) e^{j \frac{6\pi}{5} n} + (1-j2) e^{j \frac{8\pi}{5} n} \right)$$

$$= \frac{1}{5} \left(1 + ((1+j2) e^{j \frac{2\pi}{5} n} + (1-j2) e^{-j \frac{2\pi}{5} n}) + ((1-j) e^{j \frac{4\pi}{5} n} + (1+j) e^{-j \frac{4\pi}{5} n}) \right)$$

$$= \frac{1}{5} \left(1 + (e^{j \frac{2\pi}{5} n} + e^{-j \frac{2\pi}{5} n}) + j2(e^{j \frac{2\pi}{5} n} - e^{-j \frac{2\pi}{5} n}) \right. \\ \left. + (e^{j \frac{4\pi}{5} n} + e^{-j \frac{4\pi}{5} n}) - j(e^{j \frac{4\pi}{5} n} - e^{-j \frac{4\pi}{5} n}) \right)$$

$$= \frac{1}{5} \left(1 + 2 \cos\left(\frac{2\pi}{5} n\right) - 4 \sin\left(\frac{2\pi}{5} n\right) + 2 \cos\left(\frac{4\pi}{5} n\right) + 2 \sin\left(\frac{4\pi}{5} n\right) \right)$$

#

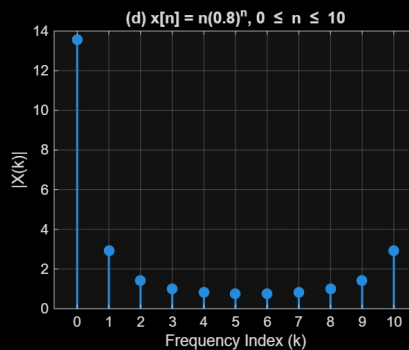
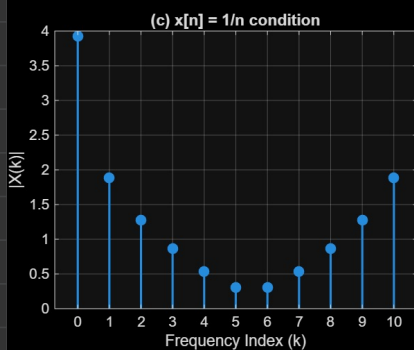
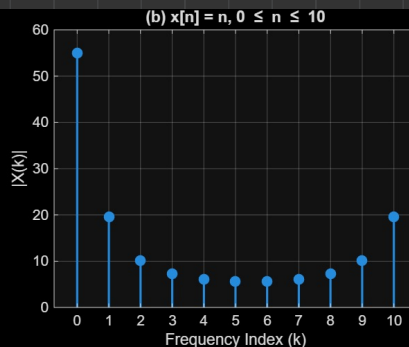
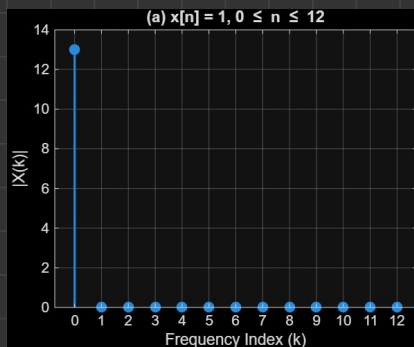
6.28 Use MATLAB to compute the FFT of the following signals. For each signal, plot $|X(k)|$.

(a) $x[n] = 1, 0 \leq n \leq 12$

(b) $x[n] = n, 0 \leq n \leq 10$

(c) $x[n] = \begin{cases} 1, & n = 0 \\ 1/n, & n = 1, 2, \dots, 10 \\ 0, & \text{otherwise} \end{cases}$

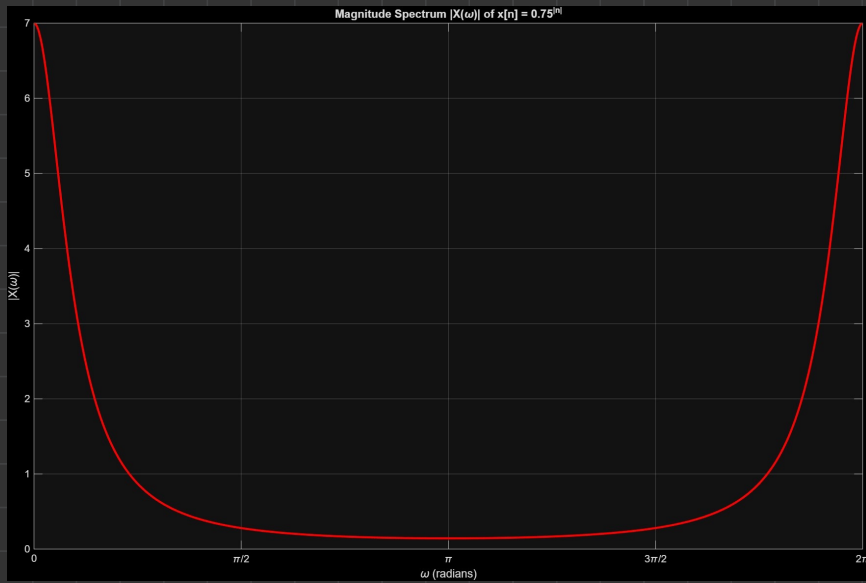
(d) $x[n] = n(0.8)^n, 0 \leq n \leq 10$



6.29 In Example 6.3, the DTFT of the signal $x[n] = a^{|n|}$ is

$$X(\Omega) = \frac{1 - a^2}{1 - 2a \cos \Omega + a^2}$$

For $a = 0.75$ and $0 < \Omega < 2\pi$, plot $|X(\Omega)|$.



6.30 Use MATLAB to find the DFT of the discrete signal

$$x[n] = \{1, 2, 0, -1, -2, 1, 5, 4\}$$

The DFT coefficients $X(k)$ are:

$$X(0) = 10.0000 + 0.0000j$$

$$X(1) = 7.2426 + 7.8284j$$

$$X(2) = -6.0000 + 0.0000j$$

$$X(3) = -1.2426 + -2.1716j$$

$$X(4) = -2.0000 + 0.0000j$$

$$X(5) = -1.2426 + 2.1716j$$

$$X(6) = -6.0000 + 0.0000j$$

$$X(7) = 7.2426 + -7.8284j$$

